READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.*

Ideal Homes

New ideas and some old ones

A The traditional kampung houses of Malaysia do not need air-conditioning. Built on stilts and with steep roofs, they have year-round ventilation. The raised structure ensures a cooling breeze comes up through the floorboards, while the high roof acts as a chimney to release hot air. The airtight, concrete boxes of modern city construction, in contrast, are heat traps, says Muhammad Peter Davis of University Putra Malaysia. He has calculated that typical modern Malaysian houses are 5°C hotter than the air outside. The builders of kampung houses "had no knowledge of modern science or engineering but they came up with the perfect design," says Davis.

B This story of ancient architectural sophistication and modern insanity is repeated around the world. In the name of modernism, people have thrown away architecture designed to cope with the environment in which they live, and adopted house designs originating from cold, northern environments, where the idea is to keep heat in.

C Once, the aim of architectural modernists was to build structures that kept nature out and to replace it with air-conditioning. Now they are learning that they cannot do that effectively. Slowly, they are seeing the benefit of working with nature, rather than against it. In California, they probably use more power for air-conditioning than anywhere else on Earth. According to Arthur Rosenfeld from the University of California, what California needs is white paint. If Los Angeles painted its roofs white, planted trees to shade buildings, and chose lighter-colored paving, it could reduce city temperatures by 3°C and cut the annual air-conditioning bill by US$170 million a year.

D Modern buildings are greedy in their use of energy. Much is made of the contribution of transportation to global warming, through its emissions of greenhouse gases. But, globally, transportation is responsible for just 22 per cent of carbon dioxide emissions. The building sector is responsible for 31 per cent, mostly the result of heating and air-conditioning systems. In developing countries, where demand for electricity for buildings is rising by 5 per cent a year, the biggest demand is for air-conditioning in modern buildings which are often designed to cope with every different climatic condition.

E Before air-conditioning, much of the Arab world kept cool through thoughtful building design. Many buildings were built according to the principles of the wind tower, a tall structure with vertical vents at the top that open in all directions to catch any passing breeze. Typically, these wind-tower buildings were made of local materials such as

stone, mud brick, wood and palm-tree fronds. The buildings were inward-looking, which served the dual function of focusing attention on the courtyard, where family members spend time together, and protecting living areas from the rays of the sun.

F Throughout the Middle East today, wind towers are often little more than museum pieces. But there are exceptions. Jordan has won awards for the architecture of a village on the shores of the Red Sea, which is designed to conserve energy. Made of rough, local stone in a traditional style, it combines the ventilation system of the wind towers for summer coolness with up-to-date heating and floor insulation to protect against the desert cold in winter.

G Traditional building materials, like traditional building designs, are being rediscovered by those looking for low-energy solutions to the current construction needs. Clay is one such material: As the Dutch housing contractor R van der Ley has argued in promoting a series of clay housing projects in developing countries, clay has many benefits over its industrialized version, brick. Clay blocks cost only half as much as ordinary bricks. Clay also generates work because people can find it, mould it, bake it and work it themselves. Two hundred clay bricks can be made with the fuel oil that makes just one ordinary brick. Moreover, clay is an excellent insulator against both cold and heat outside, and can easily be recycled.

H But although traditional methods and materials can be revived in appropriate settings, new green, low-energy technologies are needed, especially in urban environments. The thermal insulation of homes in cold countries is an example. In 1983, Sweden adopted a national standard, requiring the country's homes to lose no more than 60 kilowatt-hours per square metre over the year. To meet that standard, windows were double-glazed, and walls and roofs insulated. Every home became a fortress against the cold air outside. Unfortunately, the rest of Europe has not followed the lead.

I In the United States, Amory Lovins has promoted a range of low-energy technologies: "super-windows", for example, which let in invisible light but can be "tuned" to either allow in, or reflect away, infrared solar radiation — the stuff that heats. Buildings with large expanses of window (and big energy bills) can be designed to achieve optimum temperatures.

J As well as more efficient use of energy, the world also needs new sources of renewable power. Solar energy is being tried out in the developing world, in villages often far from electricity grids. The world's biggest solar power installation got underway in the Philippines, in 2001, where a project commenced to install solar panels for 400,000 people in 150 villages. Such projects are demonstrating that countries, whether developing or developed, no longer need huge national grids to supply electricity. Every home can do its own thing with the help of a solar panel and a storage battery.

K Houses, of course, are more than machines for living in. They are social and psychological spaces, too. Future houses may not look exactly like kampungs or wind towers or any other traditional models. But to work for us and for our environment, they will have to suit us as well as these traditional forms did.

Questions 14–18

Reading Passage 2 has eleven paragraphs, ***A–K****.*

Which paragraph contains the following information?

Write the correct letter, ***A–K****, in boxes 14–18 on your answer sheet.*

14 reasons why a particular construction material is advantageous

15 an example of a construction design which benefits domestic interaction

16 a description of a house that is ventilated from below

17 an example of self-sufficient energy supply

18 suggested methods of reducing temperatures in a city

Questions 19–22

Look at the following people (Questions 19–22) and the list of ideas below.

Match each person with the correct idea, ***A–G****.*

Write the correct letter, ***A–G****, in boxes 19–22 on your answer sheet.*

19 Muhammad Peter Davis

20 Arthur Rosenfeld

21 R van der Ley

22 Amory Lovins

List of ideas

A The choice of a certain construction material can have a socio-economic impact.

B Throughout the world, people are rejecting traditional housing design in order to appear modern.

C Houses should not only meet people's physical needs but also their social and psychological needs.

D Traditional knowledge can be superior to modern knowledge.

E There is an innovation that can save costs on both air-conditioning and heating.

F Solar energy can meet the energy needs of people living in villages in developing countries.

G There is a very simple solution that can save on the cost of air-conditioning.

Questions 24–26

Do the following statements agree with the information given in Reading Passage 2?

In boxes 24–26 on your answer sheet, write

TRUE	*if the statement agrees with the information*
FALSE	*if the statement contradicts the information*
NOT GIVEN	*if there is no information on this*

23 The air temperature in modern Malaysian houses is lower than the air temperature outside.

24 The construction industry is more to blame than transport for global warming.

25 The use of wind towers for cooling is widespread in the Middle East today.

26 The ‘super-windows’ promoted by Amory Lovins can be installed at low cost.

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Questions 14–18 | 段落信息匹配（A–K）

题号	题干翻译	答案	定位（第 X 段 + 原文 + 翻译）	详细解释
14	某种建筑材料有优势的原因	G	**第 G 段**：“**clay has many benefits over its industrialized version, brick.** Clay blocks cost only half as much… Clay also generates work… Two hundred clay bricks can be made… Moreover, clay is an excellent insulator… and can easily be recycled.” = 黏土比工业化砖材更有优势：成本更低（只要一半）、能创造工作机会、能耗更低、隔热/保温性能好、且容易回收。	题干要“为什么这种材料有优势”。第 G 段集中罗列 **clay（黏土）**的优点（成本、就业、能耗、隔热、可回收），完全对应。
15	一种能促进家庭成员互动的建筑设计例子	E	**第 E 段**：“The buildings were inward-looking… focusing attention on the courtyard, **where family members spend time together** …” = 建筑向内，注意力集中在庭院，而家人会在庭院一起度过时间。	关键词是 **domestic interaction**。只有第 E 段明确写到“家人一起相处”的空间设计（courtyard）。
16	对“从下方通风”的房屋描述	A	**第 A 段**：“The raised structure ensures **a cooling breeze comes up through the floorboards** …” = 架高结构保证凉风能从地板处向上吹入。	“from below”=从下方进风。这里明确“comes up through the floorboards”，精准对应。
17	自给自足的能源供给例子	J	**第 J 段**：“**Every home can do its own thing** with the help of a **solar panel and a storage battery**.” = 借助太阳能板和蓄电池，每个家庭都可以自给自足供能。	题干找“self-sufficient”。这句话直接总结“每家自供电”，最匹配。
18	降低城市温度的建议方法	C	**第 C 段**：“If Los Angeles **painted its roofs white**, **planted trees**… and chose **lighter-colored paving**, it could reduce city temperatures …” = 如果洛杉矶屋顶刷白、种树遮阴、用更浅色路面铺装，就能降低城市温度。	题干要求“methods”且针对“city temperatures”。第 C 段给了三条具体降温措施并说明效果。

Questions 19–22 | 人物观点配对（A–G）

题号	人名	答案	定位（第 X 段 + 原文 + 翻译）	详细解释
19	Muhammad Peter Davis	D	**第 A 段**：“The builders … **‘had no knowledge of modern science or engineering but they came up with the perfect design’** …” = 建造者不懂现代科学或工程，却想出了完美设计。	用“传统做法虽无现代工程却更好”对照现代“heat traps”，对应 **D 传统知识可能优于现代知识**。
20	Arthur Rosenfeld	G	**第 C 段**：“what California needs is **white paint**… it could… **cut the annual air-conditioning bill** …” = 加州需要白漆；刷白屋顶等能降低空调年度开支。	“very simple solution”= white paint（极简单），并明确“省空调费”，对应 **G**。
21	R van der Ley	A	**第 G 段**：“Clay also **generates work** because people can find it, mould it, bake it and work it themselves.” = 黏土还能创造工作，因为人们能自己获取、塑形、烧制并使用。	观点强调“材料选择 → 社会经济影响”，这里点明“creates jobs”，对应 **A**。
22	Amory Lovins	E	**第 I 段**：“‘**super-windows**’ … can be ‘tuned’ to either **allow in**, or **reflect away**, infrared solar radiation …” = “超级窗户”可调节：让热量进入或把热量反射出去。	既能减少过热（省空调）也能利用/保留热（省供暖），所以对应 **E**。

Questions 23–26 | TRUE / FALSE / NOT GIVEN

题号	题干翻译	答案	定位（第 X 段 + 原文 + 翻译）	详细解释
23	现代马来西亚住宅内温度比室外更低	FALSE	**第 A 段**：“typical modern Malaysian houses are **5°C hotter than the air outside**.” ＝ 典型现代马来西亚住宅比室外高 5°C。	题干说“更低”，原文说“更高”，矛盾 → FALSE。
24	建筑行业对全球变暖的责任比交通更大	TRUE	**第 D 段**：“transportation … **22 per cent** … The building sector … **31 per cent** …” ＝ 交通 22%；建筑 31%。	31%>22%，所以“更该被责怪/责任更大”成立 → TRUE。
25	如今中东普遍用风塔降温	FALSE	**第 F 段**：“wind towers are often **little more than museum pieces**. **But there are exceptions.**” ＝ 风塔通常只是博物馆展品，但也有例外。	“widespread”=普遍仍在用；原文说大多成展品，仅少数例外 → FALSE。
26	“超级窗户”安装成本低	NOT GIVEN	**第 I 段**：只讲 super-windows 的功能（调节红外热辐射、实现最佳温度），**没有任何成本/价格/安装费用**信息。	题干考“low cost”。文中未提便宜或昂贵 → NOT GIVEN。